Lakshya JEE 2.0 2025

Maths

Relations and Functions

DPP: 8

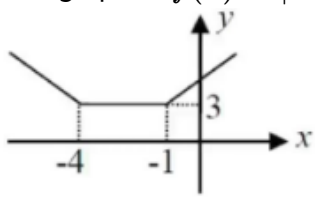
Q1 Which of following pair of function is non identical?
(A) $f(x)=|x|, g(x)=x\,\mathrm{sgn}(x)$
(B) $f(x)=\sqrt{x^2}, g(x)=|x|$
(C) $f(x)=\sqrt{x^2-6x+9}, g(x)=x-3$
(D) $f(x)=[\{x\}], g(x)=\{[x]\}$

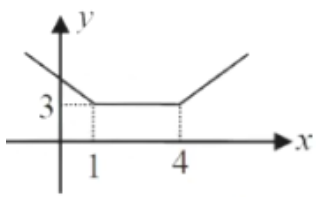
Q2 Sum of all values of 'x' satisfying $\mathrm{Max.}\{e^x, e^{-x}\}=4$ is_____

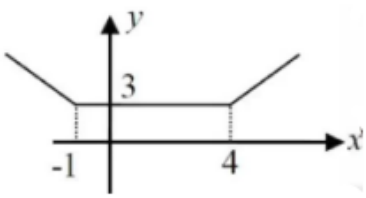
Q3 Draw graph of $f(x)=3-x^2$

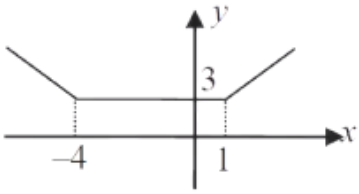
Q4 Draw graph of $f(x)=-|x+2|$

Q5 Correct graph of $f(x)=|x-1|+|x-4|$ is

(A)


(B)


(C)


(D)


Q6 If $f(x)=|x-3|-2$, then find minimum value of $f(x)$. [Hint: Draw graph]

Q7 If $f(x)=|x+3|-|5-x|$ then find range of $f(x)$. [Hint: Draw graph]

Q8 Find number of solution of following equation: $4\{x\}+[x]=2$

Q9 If $f(x)=\cos\left[\frac{\pi^2}{2}\right]x+\sin\left[\frac{-\pi^2}{2}\right]x$, $[x]$ denoting the greatest integer function, then
(A) $f(0)=0$
(B) $f\left(\frac{\pi}{3}\right)=\frac{\sqrt{3}-1}{2}$
(C) $f\left(\frac{\pi}{2}\right)=-1$
(D) $f(\pi)=0$

Q10 Number of solutions of $4\{x\}=x+[x]$ is
(A) 0 (B) 1
(C) 2 (D) 3

Q11 Solve the following system of equations
$x+[y]+\{z\}=200$
$\{x\}+y+[z]=190.1$
$[x]+\{y\}+z=178.8$

Q12 Find the set of values of x such that $[x]+[x]^2$ is prime.

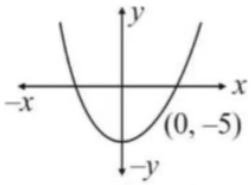
Q13 $|2x-1|=3[x]+2\{x\}$, solve for x.

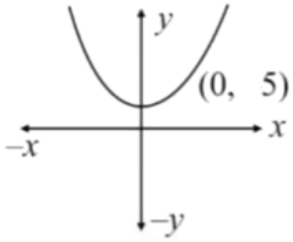
Q14 Find the value of the expression:
$\left[\frac{18}{35}\right]+\left[\frac{18(2)}{35}\right]+\left[\frac{18(3)}{35}\right]+\ldots\ldots+\left[\frac{18(33)}{35}\right]+\left[\frac{18(34)}{35}\right]$

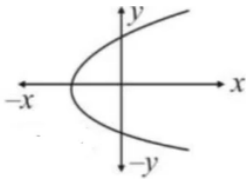
Q15

Find the value of $[\log_2(3)] + [\log_2(4)] + \ldots\ldots\ldots + [\log_2(15)]$ (where [.] denotes GIF)

Q16 The graph of $y = x^2 - 5$ is

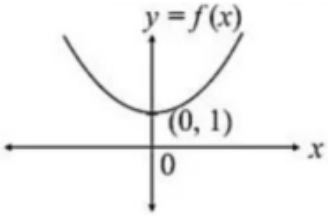
(A)


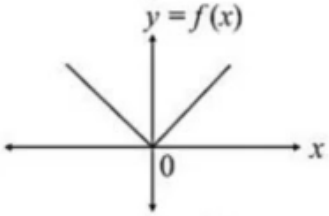
(B)


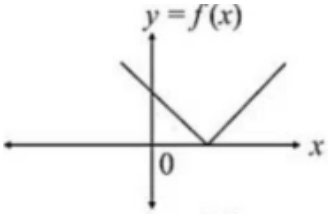
(C)


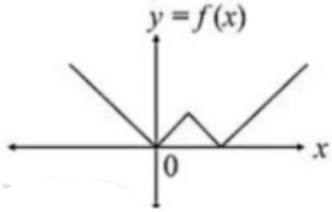
(D) None of these

Q17 Draw graph of $f(x) = \min\{|x|, |x-1|\}$

(A)


(B)


(C)


(D)


Q18 If $-1 \le [2x^2 - 3] < 2$, then x belongs to: (where [.] is G.I.F.)

(A) $-\sqrt{\frac{5}{2}} < x \le -1$ *only*

(B) $1 \le x < \sqrt{\frac{5}{2}}$ *only*

(C) $-\sqrt{\frac{5}{2}} < x \le -1$ *and* $1 \le x < \sqrt{\frac{5}{2}}$

(D) $-1 \le x \le 1$

Q19 The range of the function $f(x) = x\{x\} - x[-x]$ (where [.] and {.} denote the greatest integer function and fractional part function respectively) does not contain

(A) -1 (B) -2

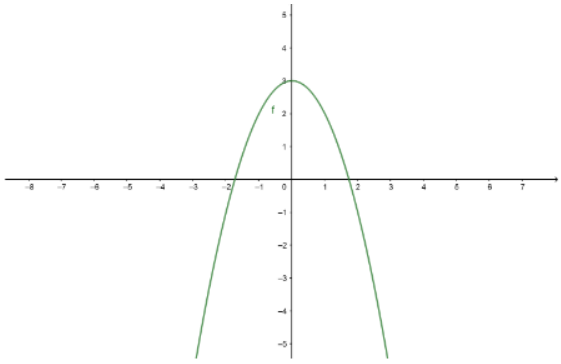
(C) 1 (D) 2

Answer Key

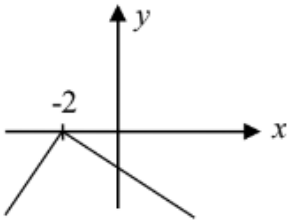
Q1 (C)

Q2 0

Q3



Q4



Q5 (B)

Q6 -2

Q7 $f(x) \in [-8, 8]$

Q8 4

Q9 (B)

Q10 (C)

Q11 x=94.65 , y=105.45, z=84.35

Q12 $x \in [-2, -1) \cup [1, 2)$

Q13 x=1/4

Q14 289

Q15 33

Q16 (A)

Q17 (D)

Q18 (C)

Q19 (A, B)